Student Name (in English): _____ Student ID: _____

PLO#K2-CLO1.1 *Asymptotic Notation* To be answered in **(10) Minutes** [ ] / 5 Marks
Question 1

**Indicate for each pair of expressions (f, g) in the table below, whether f is $O, o, \Omega, \omega, or\ \theta$ of g.** Write **'yes'** or **'no'** in each box. [4 Marks: 0.25 for each cell]

| $f$ | $g$ | $f=O(g)$ | $f=o(g)$ | $f=\Omega(g)$ | $f=\omega(g)$ | $f=\theta(g)$ |
|---|---|---|---|---|---|---|
| $f(n) = n^2 log^2 n$ | $g(n) = n^2 \cdot 2^{\log n}$ | | | | | |
| $f(n) = \log(n!)$ | $g(n) = n \log n$ | | | | | |
| $f(n) = 2^n$ | $g(n) = 3^n$ | | | | | |
| $f(n) = n^{\frac{1}{2}} \cdot \log n$ | $g(n) = n^{\frac{2}{3}}$ | | | | | |

PLO#K1-CLO 2.1 **Sorting** To be answered in **(20) Minutes** [ ] / 7 Marks
Question 2

**A. Chose the best answer. You MUST use the provided table below. Use CAPITAL letters (A, B, C, D). [4 Marks: 1 for each part]**

| 1 | 2 | 3 |
|---|---|---|
| | | |

**Que.1**: What is the order of the elements after the **third pass** of Radix sort of {bCaa, AaBb, aAab, AbcZ, Zzza, Ddcb}? Note: A < B < ... < Z < a < b < ... < z.

A. { AbcZ, bCaa, Zzza, AaBb, aAab, Ddcb}
B. {aAab, bCaa, AaBb, AbcZ , Ddcb, Zzza}
C. {AaBb, bCaa, aAab, AbcZ, Ddcb, Zzza}
D. {AaBb, AbcZ, Ddcb, Zzza, aAab, bCaa}

**Que.2**: Master method CANNOT be used when:

A. the problem is divided by a number larger than one.
B. the number of subproblems is one or more.
C. the size of the subproblems is larger than the size of the original problem.
D. All the above

**Que.3**: Which algorithm performs worst on a reverse-sorted array?

A. Insertion Sort
B. Heap Sort

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**B. For each of the following scenarios, choose the MOST suitable sorting algorithm from the following list: [4 Marks: 1 mark for each part]**

Merge Sort, Heap Sort, Counting Sort, Insertion Sort

| | Scenario | Write the Appropriate Sorting Algorithm |
|---|---|---|
| 1 | A to-do list application that usually keeps your tasks in the correct order. When you add a new task, only a couple of tasks might end up slightly out of place, while the rest of the list is already correctly sorted. Instead of re-sorting the whole list every time, the application should efficiently place the few misplaced tasks back into the right positions. | |
| 2 | A company needs to sort a very large set of employee salary records as quickly as possible. The algorithm must ensure consistent performance and maintain the order of equal elements for further sorting. | |
| 3 | A school system needs to sort hundreds of student exam scores, where each score is an integer between 0 and 100. The goal is to sort the data as quickly as possible. The systems do not have strict memory constrains allowing flexibility in choosing algorithms that may use additional space to improve performance. | |
| 4 | A company need to implement a sorting feature inside a small embedded system. The device receives incoming readings, and at certain intervals it must sort all readings as quickly as possible. The main issue is that the device has very limited RAM, so you cannot use memory-heavy algorithms that require large extra arrays. | |

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PLO#S1-CLO2.2
Question 3 *Recurrence* To be answered in **(30)** Minutes [ ] / 8 Marks

**A. Consider the following recurrence:** **[5 Marks]**

$$T(n) = \begin{cases} T\left(\frac{n}{2}\right) + T\left(\frac{n}{4}\right) + T\left(\frac{n}{4}\right) + n & n > 1 \\ \Theta(1) & n \leq 1 \end{cases}$$

**Note:** $log3_2 = 1.58, log3_4 = 0.79, log2_2 = 1,$ and $log2_4 = 0.5$

**1.** Draw the recursion tree based on the recurrence, explicitly draw at least up to level 3 and include the leaf nodes. **[1.5 Mark]**

**2.** Determine the cost at each level, including the leaf level. **[2 Marks]**

**3.** Calculate the total cost (Big-O) of T(n) of all levels. Explain your answer. **[1.5 Marks]**

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B. Based on the Master Method, complete the table below and give $\Theta(T(n))$; If the Master Method cannot be applied, write "Master Theorem does not apply" and state the reason for this. [3 Marks: 0.25 for each cell]

| $T(n)$ | $f(n)$ | $n^{\log_b a}$ | Case | $\Theta(T(n))$ |
|---|---|---|---|---|
| $T(n) = 2T\left(\frac{n}{2}\right) + n$ | | | | |
| $T(n) = 2T\left(\frac{n}{2}\right) + n^2$ | | | | |
| $T(n) = 4T\left(\frac{n}{2}\right) + n$ | | | | |

*End of Questions*
*Good Luck*